₹11,700 CRORE Estimated project cost

116 bridges

2 tunnels

3 viaducts

WHY MOUNTAIN ROADS ARE VULNERABLE



INDISCRIMINATE CUTTING

Contractors cut the hillsides steeply, beyond the accepted 45 degrees, to widen roads. Tremors generated by the drills and giant earth movers also destabilise the slopes



DRAINAGE

Heavy rains cause surface water to accumulate on the roads. This water needs to be channelised through drains and culverts to the nearby streams/rivers. For this, drains and chutes are required. Their absence leads to potholes, slope-surface erosion and heavy 'pore water pressure', reducing the 'shear strength' of the slope



DUMPING

Soil, construction waste and garbage from the city dumped on the slopes affect the vegetation while also choking the roadside drains and culverts

MITIGATION MEASURES



Geo-synthetic treatment (in which polypro-pylene and such synthetic mate-rial is used to rein-force the slopes), oil-erosion-control mats, or artificially created drainage so that the water is retained underneath the soil. Gabion walls for reinforcement and retention of slopes

The nets are capable of withstanding weights equivalent to a 25-tonne boulder falling at a speed of 100 kmph. "The active method is relatively expensive, so it is only suitable for critical areas. Passive is used where it's financially or technically impossible to stabilise the entire hill," the official added. They also showcased technology like electronic sensors along the hill slopes that trigger early warnings of landslides for motorists.

TUNNEL VISION

There is, of course, another way: tunnels. Over the past nine years, the Narendra Modi-led government at the Centre has made headlines by announcing and building large tunnels as an all-weather, much shorter alternative to commuting in the hills. The Atal Tunnel in Himachal Pradesh as a substitute to the perilous Rohtang Pass is an example. But even they contribute to hill destabilisation, no matter how sophisticated the boring machines.They are also several times more expensive.

A perusal of the Char Dham project's configuration, which, after a lengthy court battle, is widening the highways to 10 metres and double lane with paved footpaths, shows that the project engineers had factored in multiple measures to combat landslides and other calamities. It has features like steel-wire mesh for protection, and 'geo-synthetic' treatment of slopes. For this, polypropylene and such synthetic material is used to reinforce the slopes, or create soil-erosion-control mats, or even help create artificial drainage to reduce 'pore water pressure'. Now, MoRTH is working out 'mitigation measures' along stretches of the Char Dham project. On the last 100 km stretch being built by the NHIDCL, 32 locations for such measures are being identified. The solution doesn't come cheap. The mitigation measures might cost around Rs 500 crore, say officials.

But the most important factor for weather resilient mountain roads, experts say, is time. This is especially important in a young, "active" mountain range like the Himalayas. Prof. Praveen Kumar, head of the civil engineering department at IIT Roorkee, says in the past it was common practice to cut hills and let it be for a year before laying the surface of the road. "But now one can't afford to wait for too long. The cost escalates with time, the contractor doesn't find it feasible, etc. So there needs to be a balance," he says.

GOING FORWARD

"The environmental degradation we had been warning about in all our submissions is now proving to be true," says a member of Citizens for Green Doon, a civil society collective of Uttarakhand residents (they didn't want to be identified). The group was among the petitioners who had taken the central government to court to stop the mammoth highway expansion project in 2018. "The actual number of landslides is far greater than what the government records. This talk of foreign engineering solutions is just another gimmick," he says. The government, however, has time and again refuted the allegations, including in the Supreme Court, that the road project is responsible for natural disasters like landslides.

"Around 200 landslide-prone spots have been identified for rectification. On top of that, 6,000 deodar trees have been marked for felling in the Bhagirathi Eco Sensitive Zone through which a strategic road is being laid to the Indio-China border near Mana," says Dhyani of Ganga Ahwan. "All scientific studies agree that if the ecology of this zone is destabilised, it will have an adverse impact on not just Uttarakhand but all other Himalayan regions." Gadkari says he will find a lasting solution. "Roads are built in the hills because people there demand connectivity. It's a tough challenge but we are working on it," he says. Finding a balance between the two, the demands of progress and the fragile, rarefied environment of the mountains, will be key here. ■